

Machine Learning Homework 1

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Problem

Numerical computation of matrix inversion of $\mathbf{Z}^\top \mathbf{Z}$ is expensive.

Instead we could use singular value decomposition (SVD) to lower the computation cost:

$$\mathbf{Z} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$$

where:

- $\mathbf{U} = (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p)$ is an $n \times p$ orthogonal matrix
- $\mathbf{D} = \text{diag}(d_1, d_2, \dots, d_p)$ is a $p \times p$ diagonal matrix consisting of the singular values $d_1 \geq d_2 \geq \dots \geq d_p \geq 0$
- $\mathbf{V}^\top = (\mathbf{v}_1^\top, \mathbf{v}_2^\top, \dots, \mathbf{v}_p^\top)$ is a $p \times p$ matrix orthogonal matrix

Proof:

$$\begin{aligned}\hat{\beta}_\lambda^{\text{ridge}} &= (\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I}_p)^{-1} \mathbf{Z}^\top \mathbf{y} \\ &= \mathbf{V} \text{diag}_j \left(\frac{d_j}{d_j^2 + \lambda} \right) \mathbf{U}^\top \mathbf{y}\end{aligned}$$

Solution

Since $\mathbf{Z} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, we have

$$\begin{aligned}\hat{\beta}_\lambda^{\text{ridge}} &= (\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I}_p)^{-1} \mathbf{Z}^\top \mathbf{y} \\ &= ((\mathbf{U}\mathbf{D}\mathbf{V}^\top)^\top \mathbf{U}\mathbf{D}\mathbf{V}^\top + \lambda \mathbf{I}_p)^{-1} (\mathbf{U}\mathbf{D}\mathbf{V}^\top)^\top \mathbf{y} \\ &= (\mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{U}\mathbf{D}\mathbf{V}^\top + \lambda \mathbf{I}_p)^{-1} \mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{y}\end{aligned} \tag{1}$$

Note that for diagonal matrix $\mathbf{D}$, $\mathbf{D}^\top = \mathbf{D}$ and since $\mathbf{U}$, $\mathbf{V}$ are orthogonal matrices, $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_p$, $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_p$. Thus we have

$$\begin{aligned}\hat{\beta}_\lambda^{\text{ridge}} &= (\mathbf{V}\mathbf{D}^2\mathbf{V}^\top + \lambda \mathbf{I}_p)^{-1} \mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{y} \\ &= (\mathbf{V}\mathbf{D}^2\mathbf{V}^\top + \lambda \mathbf{V}^\top \mathbf{I}_p \mathbf{V})^{-1} \mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{y} \\ &= (\mathbf{V} (\mathbf{D}^2 + \lambda \mathbf{I}_p) \mathbf{V}^\top)^{-1} \mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{y}\end{aligned} \tag{2}$$

Use property $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ and property $(\mathbf{V}^\top)^{-1} = \mathbf{V}$, $\mathbf{V}^{-1} = \mathbf{V}^\top$ for orthogonal square matrix $\mathbf{V}$,

$$\begin{aligned}\hat{\beta}_\lambda^{\text{ridge}} &= (\mathbf{V}^\top)^{-1} (\mathbf{D}^2 + \lambda \mathbf{I}_p)^{-1} \mathbf{V}^{-1} \mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{y} \\ &= \mathbf{V} (\mathbf{D}^2 + \lambda \mathbf{I}_p)^{-1} \mathbf{V}^\top \mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{y} \\ &= \mathbf{V} (\mathbf{D}^2 + \lambda \mathbf{I}_p)^{-1} \mathbf{D}\mathbf{U}^\top \mathbf{y}\end{aligned} \tag{3}$$

For diagonal matrices,

$$\begin{aligned}(\mathbf{D}^2 + \lambda \mathbf{I}_p)^{-1} &= \text{diag}_j \left(\frac{1}{d_j^2 + \lambda} \right) \\ \text{diag}_j \left(\frac{1}{d_j^2 + \lambda} \right) \mathbf{D} &= \text{diag}_j \left(\frac{d_j}{d_j^2 + \lambda} \right)\end{aligned} \tag{4}$$

Finally,

$$\begin{aligned}\hat{\beta}_\lambda^{\text{ridge}} &= \mathbf{V} (\mathbf{D}^2 + \lambda \mathbf{I}_p)^{-1} \mathbf{D}\mathbf{U}^\top \mathbf{y} \\ &= \mathbf{V} \text{diag}_j \left(\frac{d_j}{d_j^2 + \lambda} \right) \mathbf{U}^\top \mathbf{y}\end{aligned} \tag{5}$$